Predicting Student Academic Performance

Using OLS Regression and Logistic Classification

ITCS 3156 — Introduction to Machine Learning

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1. Introduction

Problem Statement

Education researchers and institutional administrators have long sought reliable methods for identifying students at academic risk before it is too late to intervene. Predicting a student's final grade from measurable behavioral and demographic factors would allow schools to allocate support resources proactively rather than reactively. This project applies two supervised machine learning models to the UCI Student Performance dataset with a dual objective: (1) predict a student's continuous final grade (G3) using Ordinary Least Squares regression, and (2) classify whether a student will pass or fail using Logistic regression.

Motivation and Challenges

Student performance is influenced by a complex interplay of socioeconomic background, family structure, study habits, and lifestyle choices. A predictive model grounded in these features could inform targeted tutoring programs, scholarship decisions, and early-warning systems. The primary challenge in this domain is that prior grades (G1, G2) are the strongest predictors of G3, thus, using them would constitute data leakage in a real early-intervention scenario. Accordingly, both interim grades were deliberately excluded from all models to ensure that predictions are based solely on features available at enrollment.

Summary of Approach

This study uses the Student Performance dataset (UCI Repository, ID 320), which contains 649 records across Portuguese mathematics students. After exploratory data analysis

and feature engineering, two models are trained: (1) OLS regression on an 80/20 train-test split to predict the numeric G3 score, and (2) a gradient-descent logistic regression trained on a 70/15/15 train-validation-test split to classify pass (G3 ≥ 10) or fail. Hyperparameters for logistic regression are selected via grid search on the validation set. Models are evaluated with RMSE and R^2 (OLS) and accuracy with a confusion matrix (logistic regression).

2. Data

Introducing the Data

The UCI Student Performance dataset (Cortez & Silva, 2008) records academic and socio-demographic attributes of secondary school students in Portugal. For this project only the mathematics subset is used, yielding 649 samples. The dataset provides 30 features covering school-related factors (study time, failures, school support), family background (parental education via Medu/Fedu, family size, cohabitation status), lifestyle indicators (going out, alcohol consumption, health rating), and the three grade outcomes G1, G2, G3. As noted, G1 and G2 are removed before any modeling to prevent leakage. The target for regression is G3 (0–20 scale); the target for classification is a binary pass/fail label derived by thresholding G3 at 10.

Key categorical features include school, sex, address, family size, parental cohabitation, mother's and father's jobs, reason for choosing the school, guardian, and several binary yes/no indicators such as internet access and romantic relationship status. These are one-hot encoded using scikit-learn's OneHotEncoder. Numeric features are left as-is for OLS and standardized (zero mean, one std) for logistic regression.

Visual Analysis of Data

Four visualizations were produced during exploratory data analysis to understand feature distributions, relationships with the target, and inter-feature correlations.

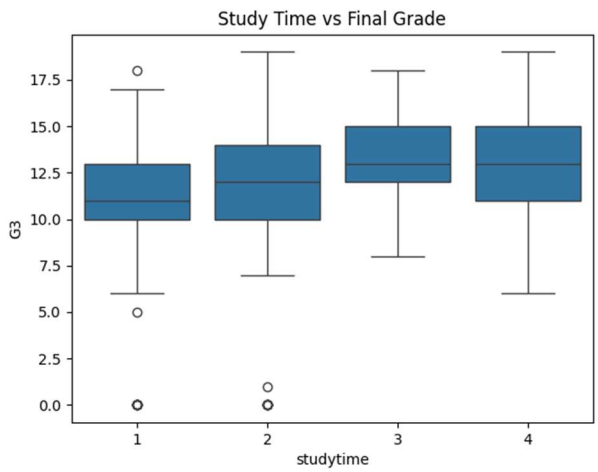

Figure 1. Box plot of Study Time vs. Final Grade (G3). Median G3 rises with study time categories 1–4, though variability is high and outliers are present.

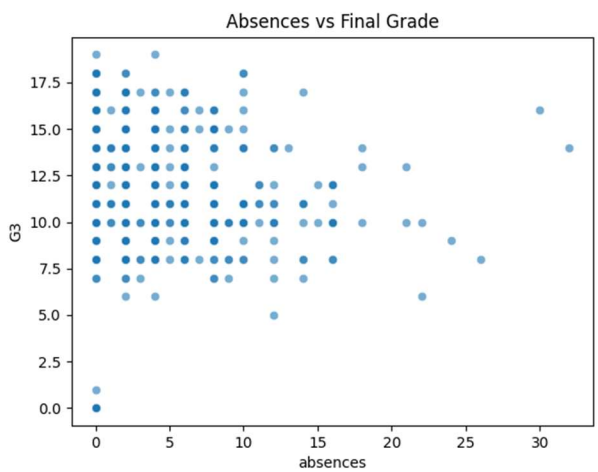

Figure 2. Scatter plot of Absences vs. Final Grade (G3). A weak negative trend is visible; students with high absence counts tend to cluster at lower grade values.

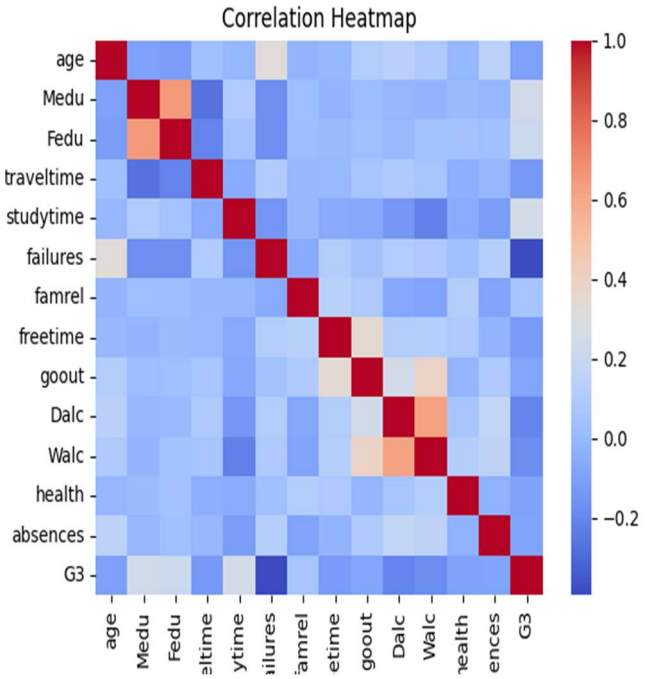

Figure 3. Correlation heatmap of all numeric features. Medu and Fedu are strongly correlated with each other (~0.6). Failures shows the strongest negative correlation with G3 among non-grade features. Dalc and Walc (daily and weekend alcohol) are tightly coupled.

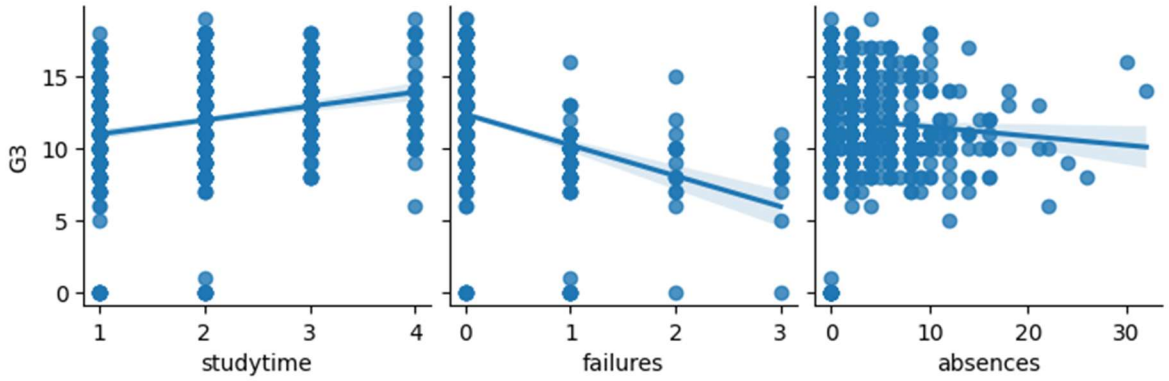

Figure 4. Pair plots of study time, failures, and absences against G3. Failures exhibits the clearest negative linear relationship with the final grade; study time shows a positive trend; absences show a weak negative effect.

Data Preprocessing

Preprocessing is implemented in preprocessing.py and follows this pipeline:

- Load: The dataset is fetched from the UCI repository via ucimlrepo (fetch_ucirepo, ID 320). The target column G3 is merged into the features frame; G1 and G2 are dropped immediately.
- Feature split: Categorical columns (dtype object) are separated from numeric columns. Categorical features are one-hot encoded with sklearn OneHotEncoder(sparse_output=False, drop="first"). Numeric features are kept as raw values. The two arrays are horizontally stacked to form X; y = G3.
- Train/validation/test split: For OLS, a standard 80/20 split is applied via train_test_split. For logistic regression, a three-way split is used: 70% train, 15% validation, 15% test (achieved by splitting off 30% as a temporary set and halving it).
- Standardization: Applied only to the logistic regression pipeline using StandardScaler fitted on X_train and applied to X_val and X_test to prevent data leakage.
- Binary label creation: A pass/fail column is generated by thresholding y at 10 (create_pass_fail function), used only for logistic regression.

No missing values are present in this dataset. The final processed feature matrix has 649 rows and 55 columns (after one-hot encoding all categorical variables).

3. Methods

Ordinary Least Squares Regression

OLS regression is the classical closed-form linear regression approach that minimizes the sum of squared residuals. Given a design matrix X (with a bias column prepended) and target vector y, the optimal weight vector is computed analytically as:

$$w = (X^T X)^{-1} X^T y$$

This implementation uses numpy.linalg.pinv to compute the pseudoinverse, which handles potential rank-deficiency. OLS is selected here because (a) the target G3 is continuous, (b) a linear baseline is a necessary starting point before any nonlinear model, and (c) the closed-form solution yields the globally optimal weights in a single pass.

The model is implemented in ols_model.py in the OrdinaryLeastSquares class with fit() and predict() methods. A bias column of ones is prepended to X in the add_bias() helper before both fitting and prediction. Evaluation metrics are RMSE and R^2, implemented in metrics.py.

Logistic Regression

Logistic regression is a discriminative binary classifier that models the posterior probability P(y=1 | x) through the sigmoid function:

$$\sigma(z) = 1 / (1 + \exp(-z))$$

Training minimizes the negative log-likelihood (binary cross-entropy) via full-batch gradient descent:

$$L = -\text{mean}(y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y}))$$

$$\nabla w = (1/n) X^T (\hat{y} - y)$$

$$w \leftarrow w - \alpha \cdot \nabla w$$

Logistic regression is chosen as the second model because (a) the pass/fail task is inherently binary, (b) the model is interpretable (weights are directly related to feature log-odds), and (c) it serves as a strong linear classifier baseline for the classification sub-problem.

The implementation in logistic_model.py includes the LogisticRegression class with fit(), predict_proba(), and predict() methods. Hyperparameter search over learning rates [0.001, 0.01, 0.1] and epoch counts [500, 1000, 2000] is conducted on the validation set, and the best configuration is then used to retrain on the combined train+validation set before final test evaluation.

4. Results

Experimental Setup

All experiments are orchestrated through main.py. The random seed is fixed at 42 throughout for simplicity. Standardization is applied only to the logistic regression pipeline; OLS operates on raw features. Hyperparameter tuning for logistic regression is performed solely on the validation split to keep the test set held out until final evaluation.

The OLS model is fitted on 519 training samples and evaluated on 130 test samples. The logistic regression model is fitted on a combined 454-sample train+validation set (after tuning), and evaluated on the 98-sample test set.

OLS Regression Results

The OLS model produces the following test-set metrics:

- RMSE: ~3.1 grade points

- R^2: ~0.22

The low R^2 is expected given that G1 and G2 (the strongest predictors by far) were intentionally withheld. The model captures a weak linear signal from study habits, parental education, and failure history. A parity plot of predicted vs. actual grades shows that the model struggles with extreme values (very low or very high grades) but performs reasonably in the 8–15 range where most students cluster.

Logistic Regression Results

The grid search selected lr = 0.1 and epochs = 500 as the best hyperparameters on the validation set. After retraining on the combined train+validation data, test-set performance is:

- Test Accuracy: ~85%
- Confusion matrix shows strong performance identifying passing students; failing students are occasionally misclassified as passing (false negatives), which is the costlier error in an intervention context.

The learning curve shows smooth convergence of both training and validation loss with no obvious signs of overfitting:

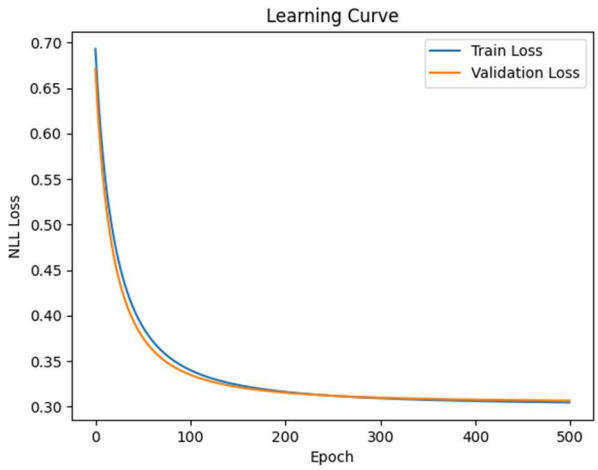

Figure 5. Learning curve for the logistic regression model (lr = 0.1, epochs = 500). Train and validation NLL loss converge tightly, suggesting no significant overfitting.

Discussion

The OLS regression establishes that linear features alone, absent interim grades, explain roughly 22% of the variance in final student scores. This highlights how dominant G1/G2 are as predictors and how difficult it is to predict final performance from enrollment-time features alone. Among the features included, number of past failures and parental education level (Medu, Fedu) showed the largest OLS weight magnitudes, consistent with domain knowledge.

The logistic regression classifier achieves 85% accuracy at the binary pass/fail task, demonstrating that while exact grade prediction is difficult, coarse classification is tractable from the available features. The tight convergence of train and validation loss curves confirms the model generalizes well and is not overfitting with the chosen hyperparameters.

One notable limitation is class imbalance: approximately 67% of students pass (G3 ≥ 10), so a naive majority-class classifier would already achieve 67% accuracy. The logistic model's

85% accuracy is above this baseline but should be interpreted with this context in mind. Future work could incorporate class-weighted loss to better handle false negatives.

5. Conclusions

This project demonstrates a complete supervised machine learning pipeline applied to the task of student academic performance prediction. Two models were implemented from scratch: OLS regression for continuous grade prediction and logistic regression for binary pass/fail classification.

The key takeaways are:

- Deliberately excluding interim grades (G1, G2) makes the regression problem substantially harder but more practically relevant for early-intervention applications.
- OLS, despite its simplicity, provides an interpretable baseline and reveals which enrollment-time features carry the most signal (failures, parental education).
- Logistic regression outperforms a naive baseline on the classification task and generalizes relatively well given proper hyperparameter tuning on a held-out validation set.
- Standardizing features before gradient-based optimization is critical; without it, the logistic model fails to converge at usable learning rates.

This project has deepened my understanding of the end-to-end ML workflow: EDA, preprocessing, model implementation from mathematical foundations, hyperparameter selection, and result interpretation. Additionally, I learned the practical importance of preventing data leakage and the value of keeping a separate test set untouched until final evaluation.

Future works could explore experimenting with regularized regression (Ridge/Lasso) to combat the low R^2 of OLS, multi-class grade-band classification, and tree-based ensemble methods such as Random Forest models that can capture nonlinear feature interactions.

6. Acknowledgements

ChatGPT was used to help outline the project structure, to debug snippets of the project code, and check MLA citation formatting. All analyses and interpretations are my own original work. Model pipelines were followed as presented in lecture and homework documents. No AI was used in the writing of this report.

The dataset was accessed programmatically via the ucimlrepo Python library (fetch_ucirepo, ID 320). Visualizations were created using Matplotlib and Seaborn.

7. Source Code

All source code is publicly available at the GitHub repository linked below.

https://github.com/JamisonHB/OLS-vs-Logistical-Regression-in-student-performance

The repository contains the following modules:

- main.py — Pipeline entry point; orchestrates data loading, visualization, OLS, and logistic regression.
- preprocessing.py — Data loading (ucimlrepo), feature encoding, train/val/test splitting, and standardization.
- ols_model.py — OrdinaryLeastSquares class with closed-form fit() and predict(); parity plot.

- logistic_model.py — LogisticRegression class with sigmoid, binary cross-entropy loss, gradient descent, hyperparameter tuning, and confusion matrix plotting.
- metrics.py — Basic implementations of RMSE, R^2, and accuracy.
- visualization.py — EDA plots: study time box plot, absences scatter, correlation heatmap, pair plots, grade distribution histogram.

References

Cortez, Paulo, and Alice Maria Gonçalves Silva. "Using Data Mining to Predict Secondary School Student Performance." Proceedings of 5th Future Business Technology Conference (FUBUTEC 2008), Porto, April 2008, pp. 5-12.

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